Study notes — PTU, *Improved regularity for a nonlocal dead-core problem*

dos Prazeres, Teymurazyan, Urbano. arXiv:2504.09557 (v2, Nov 2025). To appear, Indiana Univ. Math. J. Beginner-friendly walkthrough built for the meeting with R. Teymurazyan. Companion to `LITERATURE.md` and `PROGRESS.md`.

The equation studied:

$$-(-\Delta)^s u = u_+^\gamma - u_-^\gamma \quad \text{in } B_1, \qquad \gamma \in (0, \tfrac{1}{3}),\ s \in (\tfrac{1}{2}, 1).$$

Lesson 1 — Orientation

Physical picture (dead cores). A reactant diffuses into a catalyst pellet and is consumed by a reaction of order $\gamma < 1$. If the reaction is fast, the reactant is used up before reaching the center, leaving a region where the concentration is *exactly* zero — the **dead core**. The sublinear, non-Lipschitz term u^γ (with $0 < \gamma < 1$) is precisely what lets u reach zero on a whole set rather than at isolated points.

The equation, unpacked. - $(-\Delta)^s$ = fractional Laplacian: *nonlocal* diffusion. It averages differences $u(x) - u(y)$ over **all** of $\mathbb{R}^n$, weighted by $|x-y|^{-n-2s}$. Far-away values influence x — the root of every technical difficulty. - **Two-phase**: u has no fixed sign. RHS is u_+^γ where $u > 0$, $-u_-^\gamma$ where $u < 0$. The set where u vanishes with derivatives = **branching points** (the free-boundary analogue). - $\gamma \in (0, 1/3)$, $s \in (1/2, 1)$: forced by the method. $\gamma < 1/3 \iff \gamma < \frac{1}{1+2s}$ is the integrability condition that makes a key increment lie in the tail space L_s^1.

Main result (eq. 1.2). At a branching point x_0:

$$|u(x)| \le C\,|x - x_0|^{\frac{2s}{1-\gamma}}.$$

"Improved" because the exponent $\frac{2s}{1-\gamma} > 2s + \gamma$ beats the Schauder ceiling $C^{2s+\gamma}$.

Strategy (the whole paper in 4 moves). 1. **Scale** around the branching point: $v_r(x) = u(rx)/r^{2s/(1-\gamma)}$, study $r \to 0$. 2. **Pass to the limit** $v_r \to v_0$; the reaction term dies in the limit, leaving a clean equation. 3. **Liouville** (§3): the only such limit with the allowed growth is a polynomial of degree $\nu \in \{1, 2\}$. 4. **Contradiction**: polynomial structure forces the sharp growth (1.2).

This skeleton — *scale* $\to$ *limit* $\to$ *Liouville* $\to$ *contradict* — is reused by Correa–dos Prazeres for the p-Laplacian and is what our project ports.

Lesson 2 — The grammar of §2 (spaces, solutions, regularity toolbox)

1. The fractional Laplacian.

$$(-\Delta)^s u(x) = c_{n,s}\,\mathrm{P.V.}\int_{\mathbb{R}^n} \frac{u(x)-u(y)}{|x-y|^{n+2s}}\,dy.$$

- **P.V.**: the weight $|x-y|^{-n-2s}$ is non-integrable near $y=x$; symmetric cancellation of the odd part of $u(x)-u(y)$ makes the limit finite. Needs u locally $C^{1,1}/C^2$ at x for the classical value — motivates viscosity solutions. - **Scaling (memorize):** $(-\Delta)^s[u(\lambda\cdot)](x) = \lambda^{2s}\,((-\Delta)^s u)(\lambda x)$ — order-$2s$ derivative. This dictates the $r^{2s/(1-\gamma)}$ normalization in §4. - $c_{n,s}$: normalization, depends only on n, s; ensures $(-\Delta)^s \to -\Delta$ as $s \to 1$ (used in §5).

2. Tail space L^1_s. $\|u\|_{L^1_s} = \int_{\mathbb{R}^n} \frac{|u(y)|}{1+|y|^{n+2s}}\,dy$. Finiteness $\iff$ the far-field tail is tame enough for nonlocal diffusion to be defined/stable. **Every difficulty (the §4 gap, the p-port) reduces to whether a rescaled function stays in (the analogue of) L^1_s.**

3. Viscosity solutions (Def. 2.1). Touching-test-function notion: u is a subsolution if wherever smooth φ touches u from above at x_0, the operator inequality holds for $v=\varphi$ (in small ball) $\cup$ u (outside). Chosen because (i) never differentiates u directly, (ii) supports comparison/existence, (iii) generalizes to fully nonlinear / p-Laplacian. Note v keeps real u outside the ball — nonlocality built in.

4. Comparison principle (Thm 2.1). $Lu_1 \le 0 \le Lu_2$ in B_1 and $u_1 \ge u_2$ *outside* $B_1 \Rightarrow u_1 \ge u_2$ everywhere. Ordering on the complement (not boundary) controls the inside; powers Perron existence.

5. Hölder seminorms. $[u]_{C^\alpha} = \sup \frac{|u(x)-u(y)|}{|x-y|^\alpha}$; $[u]_{C^{1+\alpha}} = \max_{|\beta|=1}[D^\beta u]_{C^\alpha}$; $[u]_{C^{2+\alpha}}$ likewise with $|\beta|=2$. §4's θ_k just tracks these under rescaling.

6. Three regularity theorems (black boxes). - **Thm 2.2** (homogeneous): $(-\Delta)^s u = 0$, $\|u\|_{L^1_s} < \infty \Rightarrow u \in C^{1+\sigma}(B_{1/2})$, $\|u\|_{C^{1+\sigma}(B_{1/2})} \le C(\|u\|_{L^\infty(B_1)} + \|u\|_{L^1_s})$. Tail term is the price of nonlocality. **Engine for the Liouville theorem.** - **Thm 2.3** (bounded RHS): $-(-\Delta)^s u = f \in L^\infty$, $s \in (1/2, 1) \Rightarrow u \in C^{1+\sigma}(B_{1/2})$, any $\sigma < 2s-1$. *Why $s > 1/2$:* need $2s-1>0$ to be in the differentiable regime. Baseline regularity. - **Thm 2.4** (Schauder): $(-\Delta)^s u \in C^\sigma \Rightarrow u \in C^{\sigma+2s}_{loc}$. With RHS $\in C^\gamma$ gives $u \in C^{2s+\gamma}_{loc}$ = the **Schauder ceiling** the main theorem beats.

Flow: 2.3 $\to$ differentiable regime; 2.4 $\to$ ceiling $C^{2s+\gamma}$; 2.2 $\to$ engine for Liouville. Main theorem beats the ceiling at branching points ($\frac{2s}{1-\gamma}$).

Lesson 3 — The Liouville theorem (§3), the technical heart

Liouville = "entire solution + controlled growth $\Rightarrow$ polynomial."

Theorem 3.1. If $s \in (1/2, 1)$, $(-\Delta)^s(u(x+h) - u(x)) = 0$ in B_1 for all h (3.1), and $[u]_{C^{\nu+\alpha}(B_R)} \le CR^{\beta-\alpha}$ for all $R \ge 1$ with $0 \le \alpha < \beta < 1$, $\nu \in \{1, 2\}$ (3.2), then u is a degree-ν polynomial.

Why the hypothesis is on increments (3.1), not on u: that's the form the blow-up limit v_0 in §4 satisfies — v_0 isn't s-harmonic, but all its first differences are.

Proof, $\nu = 1$ (five moves): 1. **Subtract the affine part** $w = u - u(0) - Du(0) \cdot x$. Legal because (a) increments of an affine function are constant in x and $(-\Delta)^s(\text{const}) = 0$, so w solves the same increment equation — *pure linearity, the move that dies for* $p \neq 2$; (b) $[\cdot]_{C^{1+\alpha}}$ ignores affine parts so (3.2) is inherited. Now $w(0) = |Dw(0)| = 0$. 2. **Differentiate + ride growth into the tail:** $w_e = Dw \cdot e$, $w_e(0) = 0 \Rightarrow |w_e(x)| \le [w]_{C^{1,\alpha}(B_{|x|})}|x|^\alpha \le C|x|^\beta$ outside B_1. Since $\beta < 1 < 2s$, $\int_{B_1^c} |w_e|/|y|^{n+2s} < \infty \Rightarrow w_e \in L^1_s$. ($\beta < 1$ is exactly what buys this.) 3. **Stability $\Rightarrow$ w_e is s-harmonic:** w_e = lim difference quotients of w, each s-harmonic; they converge in L^1_s *globally*, so Caffarelli–Silvestre stability [11, Lem 5] passes the equation to the limit: $(-\Delta)^s w_e = 0$ in B_1. **(!) This lemma needs GLOBAL L^1_s convergence — here free, but in §4 it's exactly the gap.** 4. **Gradient-decay trick $\rho^{\beta-1} \to 0$ (transplants to p!):** $v(x) = \rho^{-\beta} w_e(\rho x)$ has $\|v\|_{L^\infty(B_1)} \le C$ uniformly; Thm 2.2 $\Rightarrow$ $\|Dv\|_{L^\infty(B_{1/2})} \le C$; but $= \rho^{1-\beta}\|Dw_e\|_{L^\infty(B_{1/2\rho})} \Rightarrow \|Dw_e\|_{L^\infty(B_{1/2\rho})} \le \rho^{\beta-1}C_3 \to 0$. So $Dw_e(0) = 0$; translation invariance $\Rightarrow$ $Dw_e \equiv 0$. 5. $w_e \equiv \text{const} = 0$ for every $e \Rightarrow Dw \equiv 0 \Rightarrow w \equiv 0 \Rightarrow$ u affine. ∎

$\nu = 2$**:** subtract quadratic Taylor poly, run the same machine on $w_{ee} \Rightarrow u$ quadratic. (This is the case §4 uses, $\beta = \frac{2s}{1-\gamma} - 2$.)

Lemma 3.1 (jet-to-growth, mean value theorem only — transplants verbatim): vanishing jet $u(0) = |D^{\nu-1}u(0)| = |D^\nu u(0)| = 0$ + $\sup_{0<r<1/2} r^{\alpha-\beta}[u]_{C^{\nu+\alpha}(B_r)} \le A$ ($\beta > \alpha$) $\Rightarrow$ $|u(x)| \le A|x|^{\nu+\beta}$ in $B_{1/2}$. Converts seminorm control + vanishing jet into pointwise growth; used in §4 to turn the contradiction hypothesis into Liouville's input.

Carry-outs for the project: 1. **Linearity used twice** (affine subtraction; derivative = limit of s-harmonic increments) — both fail for $p \neq 2$ $\Rightarrow$ must use Correa–dos Prazeres no-subtraction Liouville. 2. **Stability's global-L^1_s hypothesis** = seed of the §4 gap. 3. $\rho^{\beta-1} \to 0$ **decay trick survives** to the nonlinear/p setting.

Glossary — every recurring term, in plain language

Read this whenever a term in the notes or in `main.tex` is unfamiliar. Each entry gives the intuition first, then the precise meaning, then where it shows up.

Dead core. The interior region where the solution u is *exactly* zero. Physically: where the reactant has been completely consumed. It exists because the absorption u^γ with $\gamma < 1$ is non-Lipschitz at 0.

Branching point / free-boundary point. A point x_0 on the edge of the dead core where u touches zero *with* a horizontal tangent: $u(x_0) = 0$ and $Du(x_0) = 0$. These are the points at which the sharp growth estimate is proved. ("Free boundary" = the boundary $\partial\{u > 0\}$, unknown in advance; "branching" because the two phases $\{u > 0\}$ and $\{u < 0\}$ can meet there.)

Jet (1-jet, 2-jet). The Taylor data of u at a point. The k**-jet** is $(u(x_0), Du(x_0), \dots, D^k u(x_0))$. The **1-jet** $= (u(x_0), Du(x_0))$ is value + gradient (the best affine approximation). A **vanishing 1-jet** means $u(x_0) = |Du(x_0)| = 0$ — exactly the branching-point condition. A **vanishing 2-jet** adds $|D^2 u(x_0)| = 0$ (Hessian zero too); this is the $\nu = 2$ case, available only at $p = 2$.

ν **(nu).** The order of vanishing assumed at the branching point: $\nu = 1$ uses a vanishing 1-jet, $\nu = 2$ a vanishing 2-jet. For $p \neq 2$ only $\nu = 1$ is usable (quadratics aren't (s, p)-harmonic).

Fractional Laplacian $(-\Delta)^s$**.** Nonlocal diffusion of order $2s$. $(-\Delta)^s u(x) = c_{n,s}\,\mathrm{P.V.}\int \frac{u(x)-u(y)}{|x-y|^{n+2s}} dy$ — averages how $u(x)$ differs from u everywhere else.

Fractional p**-Laplacian** $(-\Delta_p)^s$**.** The nonlinear ($p \neq 2$) version: replace the difference $u(x) - u(y)$ by $J_p(u(x) - u(y)) = |u(x) - u(y)|^{p-2}(u(x) - u(y))$. Reduces to $(-\Delta)^s$ when $p = 2$.

s**-harmonic /** (s, p)**-harmonic.** A function with $(-\Delta)^s u = 0$ (resp. $(-\Delta_p)^s u = 0$) — the nonlocal analogue of "harmonic" ($\Delta u = 0$). These are the "trivial" solutions the Liouville theorem classifies.

P.V. (principal value). A symmetric way to make sense of the singular integral defining the operator (the weight $|x - y|^{-n-2s}$ blows up at $y = x$); the odd part cancels, leaving a finite value.

Tail / L^1_s **/** $\mathrm{Tail}_{p-1,sp}$**.** Bookkeeping for the far-field. Because the operator is nonlocal, it only makes sense if u doesn't grow too fast at infinity. $\|u\|_{L^1_s} = \int \frac{|u(y)|}{1+|y|^{n+2s}} dy$ (the $p = 2$ version); $\mathrm{Tail}_{p-1,sp}$ is its p-analogue. "Finite tail" = "far-field tame enough for the theory to work." Almost every difficulty in the project is a tail question.

Seminorm $[u]_{C^\alpha}$**, Hölder space** $C^{k,\alpha}$**.** Measures of smoothness. $[u]_{C^\alpha} = \sup_{x \neq y} \frac{|u(x)-u(y)|}{|x-y|^\alpha}$ quantifies α-Hölder continuity; $C^{k,\alpha} = C^{k+\alpha}$ means k derivatives, the k-th being α-Hölder. $[u]_{C^{1+\alpha}}$ controls the gradient's Hölder norm.

Viscosity solution. A notion of solution needing no derivatives of u: test against smooth functions touching u from above/below. Used because solutions may not be twice differentiable, and because it extends to nonlinear/p operators.

Weak solution. The other notion of solution, defined by integrating against test functions ($u \in W^{s,p}$). For this problem weak $\equiv$ viscosity (Korvenpää–Kuusi–Lindgren).

Blow-up / rescaling. Zoom in at a branching point: $v_r(x) = u(rx)/r^\alpha$, study $r \to 0$. The exponent α is chosen so the rescaled functions stay balanced; their limit v_0 is a clean model solution.

Homogeneity (of the operator). $(-\Delta_p)^s(\lambda u) = \lambda^{p-1}(-\Delta_p)^s u$ for $\lambda > 0$ ($(p-1)$-homogeneity), and the order-sp spatial scaling. In our project these two together make the absorption term vanish along the blow-up — *this replaces the differentiation step that needs linearity*.

Liouville theorem. "Entire solution + controlled growth ⇒ trivial (a polynomial / zero)." The engine that classifies the blow-up limit v_0 and forces the contradiction. The project's "load-bearing wall."

Increment / first difference w^h. $w^h(x) = u(x+h) - u(x)$. Differencing lowers the growth order by one and, for $p \neq 2$, turns the nonlinear equation into a *linear* one for w^h (with a u-dependent kernel). **Second difference** $\Delta_h^2 u = u(\cdot{+}2h) - 2u(\cdot{+}h) + u(\cdot)$ lowers it by two.

Cocycle identity. The bookkeeping rule $w^{h_1+h_2} = w^{h_1}(\cdot + h_2) + w^{h_2}$ for increments; used to show a quantity depending on h is linear in h.

Stability lemma. The theorem letting you pass an equation to the limit: if $u_k \to u$ (with global tail control) and $(-\Delta)^s u_k = f_k$, then $(-\Delta)^s u = \lim f_k$. The hypothesis that the convergence be *global* (in L^1_s) is the subtle one — the source of the §4 gap.

θ_k **device.** The contradiction machine (PTU §4 / CP §6): assume the estimate fails, build a sequence whose rescaled blow-up survives a normalization but must vanish by Liouville. $\theta_k(r')$ measures the worst seminorm growth across scales.

Gradient anchor. A trick ($\int \eta\, D_e v = \int D_e \eta\, v \leq C$) that pins the gradient at one point, giving local bounds on v_k *without* normalizing it in L^∞ — useful nonlocally.

Schauder estimate / "better than Schauder." Schauder theory gives the generic a-priori regularity ($C^{2s+\gamma}$ for $p = 2$). "Better than Schauder" = the growth at branching points is *smoother* than that generic ceiling — the whole point of the word "improved" in the title.

Nondegeneracy. Two uses. (1) Lower bound matching the growth: $\sup_{B_r} |u| \geq c\, r^\alpha$ (makes the estimate sharp). (2) For the $p \neq 2$ Liouville: that the limit v_0 has $Dv_0 \not\equiv 0$ on open sets, so the increment's v_0-dependent kernel stays uniformly elliptic.

Admissible region. The set of parameters (s, p, γ) for which the theorem's hypotheses hold simultaneously — the analogue of PTU's $\gamma \in (0, 1/3), s \in (1/2, 1)$. Determined by intersecting the constraint conditions (C1 tail, C2 the cap $\alpha < 2$, etc.).